

FINAL MARK

**GIRRAWEE HIGH SCHOOL
MATHEMATICS
YEAR 11 HSC TASK 1 2009
ANSWERS COVER SHEET**

Name: _____

QUESTION	MARK	H2	H3	H4	H5	H6	H7	H8	H9
Q1	/12			✓	✓				✓
Q2	/14			✓	✓				✓
Q3	/14			✓	✓				✓
Q4	/17	✓		✓	✓				✓
Q5	/24				✓				✓
Q6	/22				✓				✓
Q7	/18				✓				✓
TOTAL	/121	/17		/57	/121				/121

HSC Outcomes

Mathematics

- H2 constructs arguments to prove and justify results.
- H3 manipulates algebraic expressions involving logarithmic and exponential functions.
- H4 expresses practical problems in mathematical terms based on simple given models.
- H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems.
- H6 uses the derivative to determine the features of the graph of a function.
- H7 uses the features of a graph to deduce information about the derivative.
- H8 uses techniques of integration to calculate areas and volumes.
- H9 communicates using mathematical language, notation, diagrams and graphs.

GIRRAWEE HIGH SCHOOL MATHEMATICS

YEAR 11

HSC Task 1, 2009

Time Allowed: 90 minutes

Instructions: Attempt all questions.
Start each question on a new page.
All necessary working must be shown.
Marks may be deducted for careless or badly arranged work.

Question 1 (12 marks)

- a) A child's printing set contains five lots of each of the letters of the alphabet. What is the probability of reaching into a box containing all the letters and choosing:
- i) the letter W? 1
 - ii) a vowel? 1
 - iii) a consonant? 2
- b) In a game, tiles in a bag are marked 1, 2 or 3. There are equal quantities of each. A tile is removed, noted and replaced.
- i) Construct a Probability Tree diagram showing the outcomes for two successive draws. 3
 - ii) Calculate the probability of drawing:
 - α) two 3's 1
 - β) no 3's 2
 - γ) at least one three 2

Question 2 (14 marks)

- a) Find the locus of the point $P(x, y)$ that moves so that:
- i) it is equidistant from the points $A(-1, 6)$ and $B(3, 2)$. 3
 - ii) its distance from $A(2, 4)$ is always twice its distance from $B(3, -2)$. 4
 - iii) its distance from the y -axis is equal to its distance from the point $(2, 3)$. 5
 - iv) it is 3 units from the point $(3, 6)$. 2

Question 3 (14 marks)

a) Find the focus and directrix of:

i) $x^2 = 2y$ 3

ii) $y^2 = -4x$ 3

b) Find the equation of the parabola with its focus at (1, 4) and directrix at $y = 8$. 3

c) A parabola has its vertex at the point (3, 1) and focus at the point (3, 3).

i) What is the focal length? 1

ii) What is the equation of the directrix? 2

iii) What is the equation of the parabola? 2

Question 4 (17 marks)

a) Sketch the graph $y^2 = x$ marking its focus and directrix. 3

b) Find the radius and coordinates for the centre of the circle
 $x^2 + y^2 - 4x - 10y + 20 = 0$. 3

c) Find the equations of the *tangent* and *normal* to the parabola $x^2 = -8y$
at point (4,-2). 6

d) Points $P(8,8)$ and $Q\left(-2, \frac{1}{2}\right)$ are on the parabola $x^2 = 8y$.
Show that PQ is a focal chord. 5

Question 5 (30 marks)

- a) For the series $5+12+19+26+\dots$, find
- i) the common difference. 1
 - ii) the 24th term. 2
 - iii) the sum of the first 24 terms. 2
- b) If the seventh term of an arithmetic sequence is 20 and the thirteenth term is 38, find the first term and the common difference. 4
- c) Find the number of terms in an arithmetic sequence with $a=5$, $d=2$ and the last term 43. 2
- d) For the series $12+6+3+\dots$, find
- i) the common ratio. 1
 - ii) the 6th term. 2
 - iii) the sum of the first 6 terms. 2
- e) The third term of a geometric sequence is 8, the sixth term is -1, find
- i) the common ratio. 4
 - ii) the first term. 2
 - iii) the eighth term. 2

Question 6 (22 marks)

- a) Find whether 2863 is a term of the sequence $5, 8, 11, \dots$ 3
- b) The numbers $2, a, b$ are in arithmetic progression and $a, b, 9$ are in geometric progression. Find a and b . 6
- c) Express $0.5\overline{7}$ as a fraction, using a geometric series. 3
- d) Find the common ratio of a geometric series with a first term of 125 and a limiting sum of 100. 3
- e) If the sequence $4, -2.5, \dots$ is geometric, find:
- i) an expression for the n th term. 2
 - ii) an expression for the sum of n terms and find its value when $n=5$. 3
 - iii) the limit of S_n as $n \rightarrow \infty$. 2

Question 7 (18 marks)

a) Evaluate:

i) $\sum_{n=1}^5 (-5)^{n-1}$	2	ii) $\sum_{n=10}^{20} (15 - n)$	2
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b) In a class of 24 students 3 play no sport, 14 play cricket and 12 play tennis. If a student is chosen at random;

- | | | |
|------|--|---|
| i) | Draw a venn diagram to represent the data. | 2 |
| ii) | What is the probability that he or she plays tennis only? | 1 |
| iii) | What is the probability that he or she plays both tennis and cricket? | 1 |
| iv) | If two students are chosen at random, which is more likely: They both play cricket only or they both play tennis only. | 3 |

c) Cans of fruit in a supermarket display are stacked so that there are 4 cans in the top row, 6 in the next row, 8 in the next and so on. If there are 10 rows in the display, find:

- | | | |
|-----|--|---|
| i) | the number of cans in the bottom row. | 2 |
| ii) | the total number of cans in the display. | 2 |

d) The lengths of the rungs of a ladder increase uniformly from 40cm in the top rung to 75cm in the bottom rung. If 13.8m of wood are used to make the rungs, how many rungs are there?

3

Question 1.

a) i) $P(W) = \frac{1}{26}$

(1)

ii) $P(\text{Vowel}) = \frac{5}{26}$

(1)

iii) $P(\text{consonant}) = 1 - P(\text{Vowel})$
 $= \frac{21}{26}$

(2)

(3) bi) 1st 2nd Outcome P

$\frac{1}{3}$	1	11	$\frac{1}{9}$
$\frac{1}{3}$	2	12	$\frac{1}{9}$
$\frac{1}{3}$	3	13	$\frac{1}{9}$
$\frac{1}{3}$	1	21	$\frac{1}{9}$
$\frac{1}{3}$	2	22	$\frac{1}{9}$
$\frac{1}{3}$	3	23	$\frac{1}{9}$
$\frac{1}{3}$	1	31	$\frac{1}{9}$
$\frac{1}{3}$	2	32	$\frac{1}{9}$
$\frac{1}{3}$	3	33	$\frac{1}{9}$

ii) $P(\text{two 3's}) = \frac{1}{3} \times \frac{1}{3}$

$= \frac{1}{9}$

(1)

iii) $P(\text{no 3's}) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{1}{3}$

$= \frac{4}{9}$

(2)

iv) $P(\text{at least one 3})$

$= 1 - P(\text{no 3's})$

$= 1 - \frac{4}{9} = \frac{5}{9}$

(2)

Question 2

i) $A(-1, 6)$ $B(3, 2)$ $P(x, y)$

$PA = PB$

$PA^2 = PB^2$

$(x+1)^2 + (y-6)^2 = (x-3)^2 + (y-2)^2$

$x^2 + 2x + 1 + y^2 - 12y + 36 = x^2 - 6x + 9 + y^2 - 4y + 4$

$8x - 8y + 24 = 0$

$\therefore x - y + 3 = 0$

(3)

ii) $A(2, 4)$ $B(3, -2)$ $P(x, y)$

$PA = 2 \times PB$

$PA^2 = 4 \times PB^2$

$(x-2)^2 + (y-4)^2 = 4[(x-3)^2 + (y+2)^2]$

$x^2 - 4x + 4 + y^2 - 8y + 16 = 4[x^2 - 6x + 9 + y^2 + 4y + 4]$

$x^2 - 4x + 4 + y^2 - 8y + 16 = 4x^2 - 24x + 36 + 4y^2 + 16y + 16$

$3x^2 - 20x + 3y^2 + 24y + 32 = 0$

(4)

iii) point $(2, 3)$ y -axis $P(x, y)$

$A(2, 3)$ $B(0, y)$ $P(x, y)$

$PA = PB$

$PA^2 = PB^2$

$(x-2)^2 + (y-3)^2 = (x-0)^2 + (y-y)^2$

$x^2 - 4x + 4 + y^2 - 6y + 9 = x^2$

$y^2 - 6y - 4x + 13 = 0$

$y^2 - 6y + 9 = 4x - 4$

$(y-3)^2 = 4x - 4$

$(y-3)^2 = 4(x-1)$

(5)

OR

$(y-k)^2 = 4a(x-h)$

$(y-3)^2 = 4(1)(x-1)$

$(y-3)^2 = 4(x-1)$

iv) Point $(3, 6)$ $P(x, y)$

$PA = 3$

$PA^2 = 9$

$(x-3)^2 + (y-6)^2 = 9$

ci) $V(3, 1)$ $S(3, 3)$
 $a = 2$

(1)

ii) directrix

$y = -1$

(2)

iii) $(x-3)^2 = 4(2)(y-1)$

$(x-3)^2 = 8(y-1)$

(2)

Question 4

a) $y^2 = x$

$4a = 1$

$a = \frac{1}{4}$

$S(\frac{1}{4}, 0)$

directrix: $x = -\frac{1}{4}$

Question 3

ai) $x^2 = 2y$

$x^2 = 4ay$

$4a = 2$

$a = \frac{1}{2}$

$\therefore S(0, \frac{1}{2})$

directrix: $y = -\frac{1}{2}$

(3)

ii) $y^2 = -4x$

$y^2 = -4ax$

$4a = 4$

$a = 1$

$\therefore S(-1, 0)$

directrix: $x = 1$

(3)

b) $(x-1)^2 = 4(2)(y-6)$

$(x-1)^2 = 8(y-6)$

(3)

b) $x^2 + y^2 - 4x - 10y + 20 = 0$

$x^2 - 4x + 4 + y^2 - 10y + 25 = -20 + 4 + 25$

$(x-2)^2 + (y-5)^2 = 9$

$\therefore$ Centre of the circle is $(2, 5)$ and the radius = 3

(3)

c) $x^2 = -8y$ $(4, -2)$

$y = -\frac{x^2}{8}$

$y' = -\frac{2x}{8} = -\frac{x}{4}$

When $x = 4$

$y' = -\frac{4}{4} = -1$

$m_{\text{tangent}} = -1$ $m_{\text{normal}} = 1$

Eqn normal: $y+2 = 1(x-4)$
 $y+2 = x-4$
 $y = x-6$ or
 $x-y-6 = 0$

Eqn tangent: $y+2 = -1(x-4)$
 $y+2 = -x+4$
 $y = 2-x$ or
 $x+y-2 = 0$

d) $xc^2 = 8y$
 $4a = 8$
 $a = 2$
 $\therefore S(0, 2)$

$m_{PQ} = \frac{\frac{1}{2}-8}{-2-8}$
 $= -7\frac{1}{2}$
 -10
 $= \frac{15}{20} = \frac{3}{4}$

Eqn PQ: $y-8 = \frac{3}{4}(x-8)$
 $4y-32 = 3x-24$
 $3x-4y+8 = 0$

When $S(0, 2)$

$3(0) - 4(2) + 8 = 0$
 $0 - 8 + 8 = 0$
 $0 = 0$

$\therefore$ The chord PQ passes through the focus so it is a focal chord.

Question 5.

a) $12-5 = 7$
 $\therefore d = 7$

ii) $a = 5$ $d = 7$
 $T_{24} = 5 + (24-1)7$
 $= 5 + 23(7)$
 $= 166$

iii) $S_n = \frac{n}{2} [2a + (n-1)d]$
 $S_{24} = \frac{24}{2} [10 + 23(7)]$

$= 12(10 + 161)$
 $= 2052$

b) $T_7 = 20$ $T_{13} = 38$
 $20 = a + 6d$ $38 = a + 12d$
Solve simultaneously.
 $20 = a + 6d$
 $38 = a + 12d$
 $18 = 6d$
 $d = 3$

Sub $d = 3$ into.
 $20 = a + 6d$
 $20 = a + 18$
 $\therefore a = 2$

$\therefore$ First term is 2 common diff 3.

c) $a = 5$, $d = 2$, $l = 43$.

$T_n = 43$
 $43 = 5 + (n-1)2$
 $43 = 5 + 2n - 2$
 $40 = 2n$
 $\therefore n = 20$

d) $r = \frac{b}{12} = \frac{1}{2}$

ii) $T_n = ar^{n-1}$ $a = 12$ $r = \frac{1}{2}$
 $T_6 = 12\left(\frac{1}{2}\right)^5$
 $= \frac{3}{8}$

iii) $S_n = \frac{a(1-r^n)}{1-r}$
 $S_6 = \frac{12\left(1 - \left(\frac{1}{2}\right)^6\right)}{\frac{1}{2}}$
 $= \frac{189}{8} = 23\frac{5}{8}$

e) $T_3 = 8$ $T_6 = -1$
i) $8 = ar^2$ $-1 = ar^5$
Solving simultaneously.
 $-1 = ar^5$
 $8 = ar^2$
 $-\frac{1}{8} = r^3$

$\therefore r = -\frac{1}{2}$

ii) $8 = a\left(-\frac{1}{2}\right)^2$
 $8 = \frac{a}{4}$
 $\therefore a = 32$

iii) $T_8 = 32\left(-\frac{1}{2}\right)^7$
 $= 32 \times -\frac{1}{128}$
 $= -\frac{1}{4}$

Question 6

a) $a = 5$ $d = 3$
 $2863 = 5 + (n-1)3$
 $2863 = 5 + 3n - 3$
 $2861 = 3n$

$\therefore n = 953.667$
 $\therefore 2863$ is not a term in this sequence.

b) AP GP.
 $a-2 = b-a$ $b = 9$
 $b = 2a-2$ a b

$\therefore 2a-2 = 9$
 a $2a-2$

$(2a-2)^2 = 9a$
 $4a^2 - 8a + 4 = 9a$
 $4a^2 - 17a + 4 = 0$
 $4a^2 - 16a - a + 4 = 0$
 $4a(a-4) - (a-4) = 0$
 $(4a-1)(a-4) = 0$
 $a = \frac{1}{4}$ and $a = 4$

$$\text{Sub } a = \frac{1}{4} \quad d = 4$$

$$b = 2\left(\frac{1}{4}\right) - 2 \quad b = 2(4) - 2$$

$$b = -1\frac{1}{2} \quad b = 6$$

$$\therefore a = \frac{1}{4} \quad \text{or} \quad a = 4$$

$$b = -\frac{3}{2} \quad b = 6$$

(b)

c)

$$0.57 = 0.57 + 0.0057 + 0.000057$$

$$a = 0.57$$

$$r = 0.01$$

$$S_{\infty} = \frac{0.57}{1 - 0.01}$$

$$= \frac{0.57}{0.99} = \frac{57}{99}$$

$$= \frac{19}{33} \quad (3)$$

$$d) a = 125 \quad S_{\infty} = 100$$

$$100 = \frac{125}{1 - r}$$

$$100(1 - r) = 125$$

$$100 - 100r = 125$$

$$-100r = 25$$

$$r = -\frac{1}{4}$$

$$\therefore \text{Common ratio} = -\frac{1}{4}$$

$$e) i) a = 4 \quad r = -0.625$$

$$T_n = 4(-0.625)^{n-1} \quad (2)$$

$$ii) S_n = \frac{4(1 - (-0.625)^n)}{1 + 0.625}$$

$$= \frac{4 - 4(-0.625)^n}{1.625}$$

When $n = 5$.

$$S_5 = \frac{4 - 4(-0.625)^5}{1.625} \quad (3)$$

$$= 2.7 \quad (2dp)$$

$$iii) S_{\infty} = \frac{4}{1 + 0.625}$$

$$= 2.46 \quad (2dp) \quad (2)$$

Question 7.

a) i)

$$\sum_{n=1}^5 (-5)^{n-1} = 1 - 5 + 25 - 125 + 625$$

$$= 521 \quad (2)$$

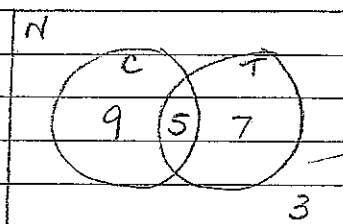
$$ii) \sum_{n=10}^{20} (15 - n)$$

$$n=10$$

$$= 5 + 4 + 3 + 2 + 1 + 0 - 1 - 2 - 3 - 4 - 5$$

$$= 0 \quad (2)$$

b) i)



$$ii) P(\text{Tennis only}) = \frac{7}{24} \quad (1)$$

$$iii) P(\text{Both Tennis and Cricket}) = \frac{5}{24} \quad (1)$$

$$iv) P(\text{Both Cricket only}) = \frac{9 \times 8}{24 \times 23} = \frac{72}{552}$$

$$P(\text{Both Tennis only}) = \frac{7 \times 6}{24 \times 23}$$

$$= \frac{42}{552}$$

$$= \frac{7}{92} \quad (3)$$

$\therefore$ It is more likely they both play cricket only.

$$c) i) a = 4 \quad d = 2 \quad n = 10$$

$$T_{10} = 4 + (10-1)(2) = 4 + (9)(2) = 22 \quad (2)$$

$\therefore$ There are 22 cars in the bottom row.

$$ii) S_{10} = \frac{10}{2} [2(4) + (10-1)(2)]$$

$$= 5(8 + 18)$$

$$= 5 \times 26$$

$$= 130 \quad (2)$$

$\therefore$ There are 130 cars in the display.

$$d) a = 40 \quad L = 75$$

$$S_n = 1380$$

$$1380 = \frac{n}{2} (40 + 75)$$

$$2760 = 115n$$

$$n = \frac{2760}{115}$$

$$n = 24$$

$\therefore$ There are 24 rungs on the ladder.